

Cloud Grundlagen und Programmierung

Wahlmodul Sommersemester

Raphael Herding

Ablauf

- Vorlesung zur Grundlagenvermittlung
- Vertiefung einzelner Themen durch die Teilnehmer
- Entwicklung und Test eines Prototypen zur Veranschaulichung der erlernten Konzepte
- Modulprüfung:
 - Abschlussdokumentation
 - Präsentation

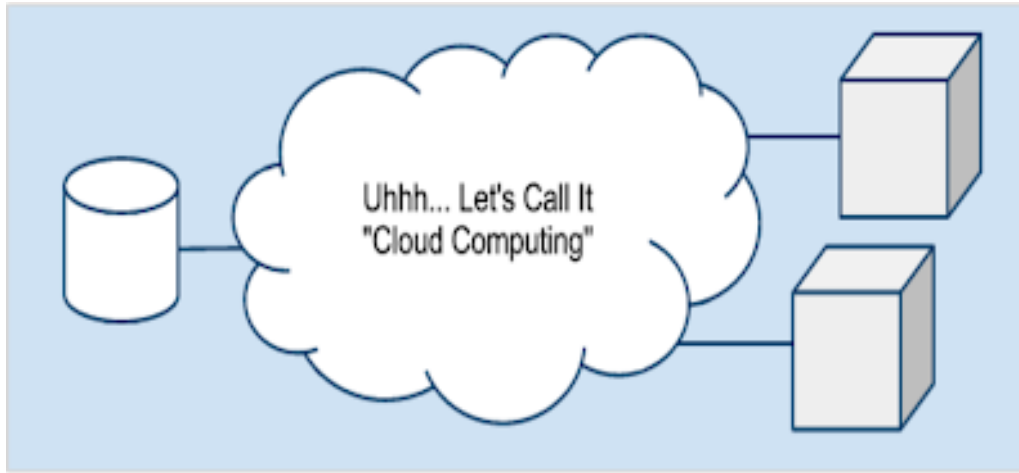
Sources

Reference to further reading:

<https://data-flair.training/blogs/cloud-computing-tutorial/>

Fundamentals

Fundamentals – Concepts and Definitions



- The name cloud computing was inspired by the cloud symbol of network flow chart diagrams.
- It is a general term for anything that involves delivering hosted service over the Internet.
- The word cloud was used as a metaphor for the Internet.

"Cloud" comes from the early days of Internet where we drew the network as a cloud... we didn't care where the message went- the cloud hid it from us" , Kevin Marks, Google

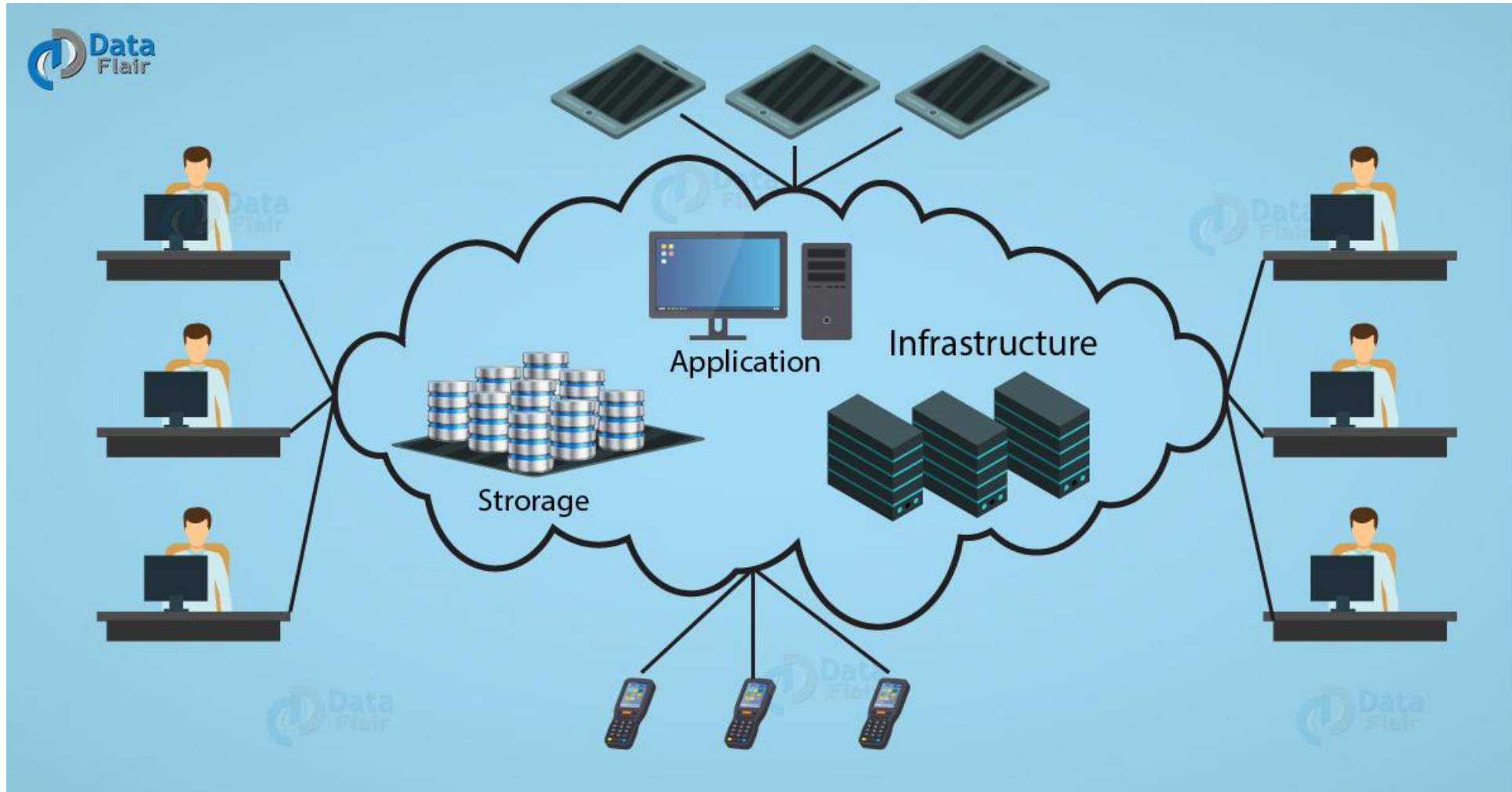
Fundamentals – Concepts and Definitions

- Cloud computing is a **service**, which offers consumers to work over the internet.
- Storing and accessing data and programs over the internet rather than the computer's hard disk.
- The user can access the data from anywhere.
- In most cases, the user needs to have a valid registered user id.

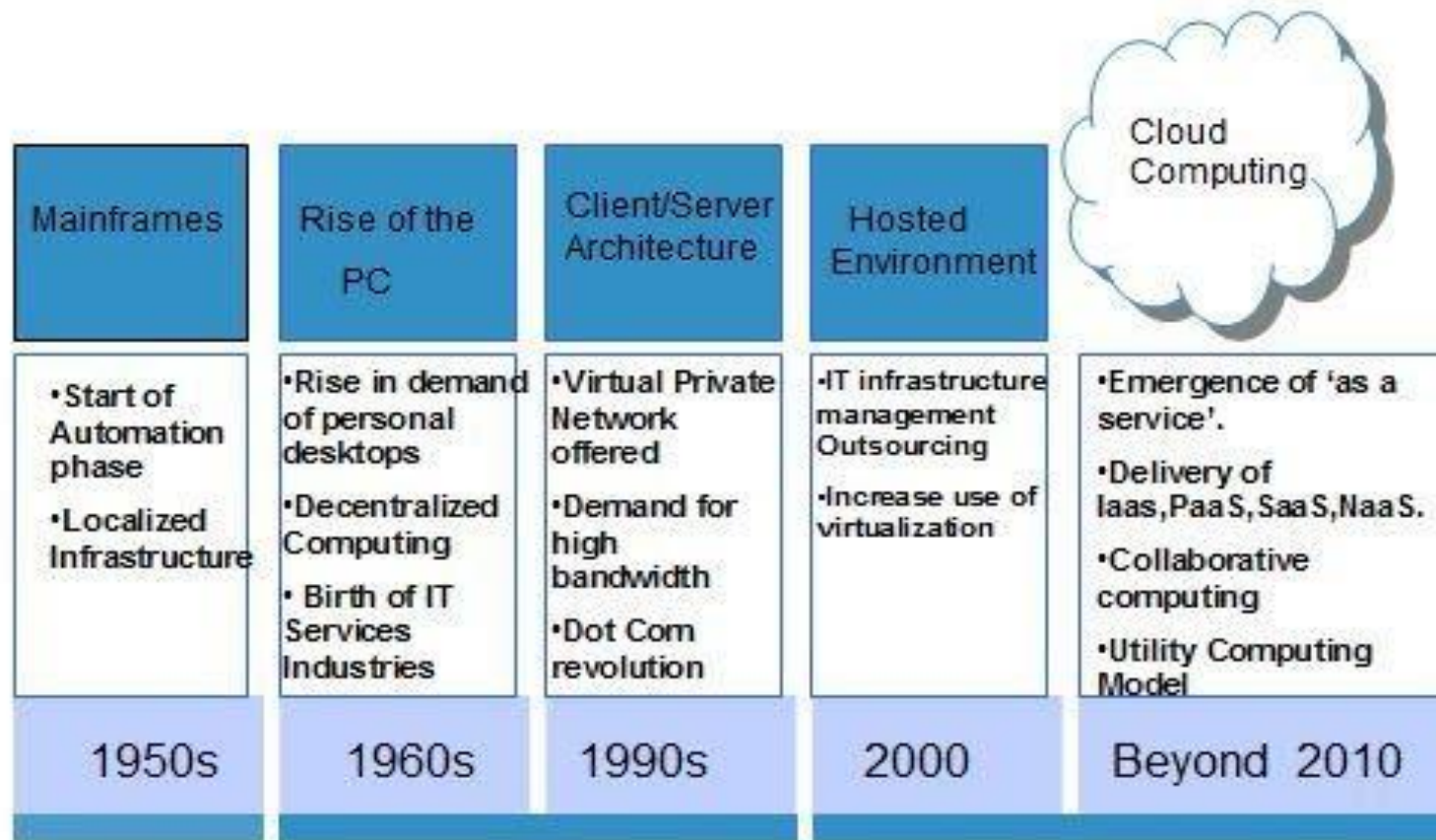
- The „Cloud“ is operated and managed by the cloud **provider**.
- The provider offers a high amount of resources to their **consumers**.

- Goal: Consumers can focus on his key competency.

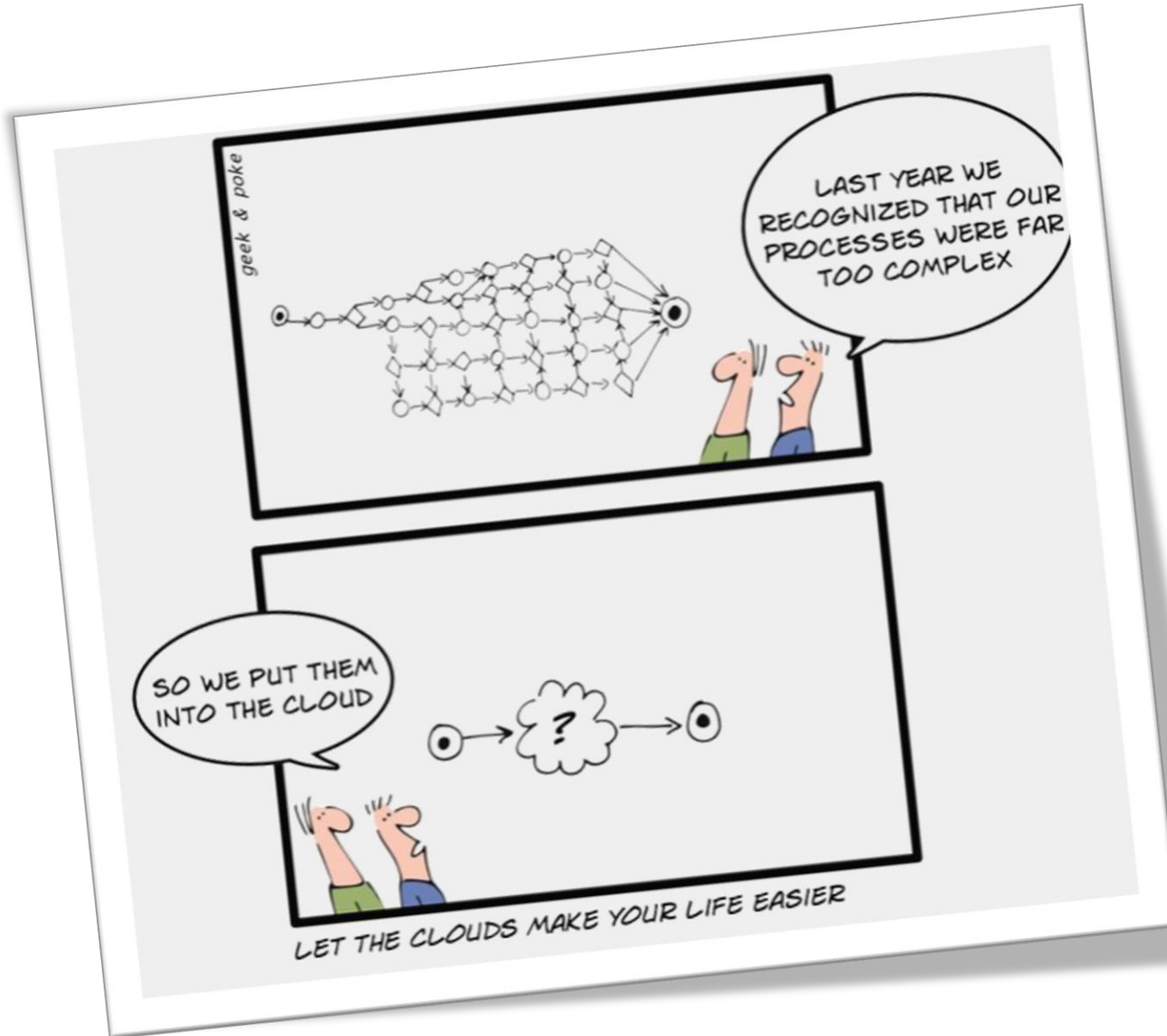
Fundamentals – Concepts and Definitions



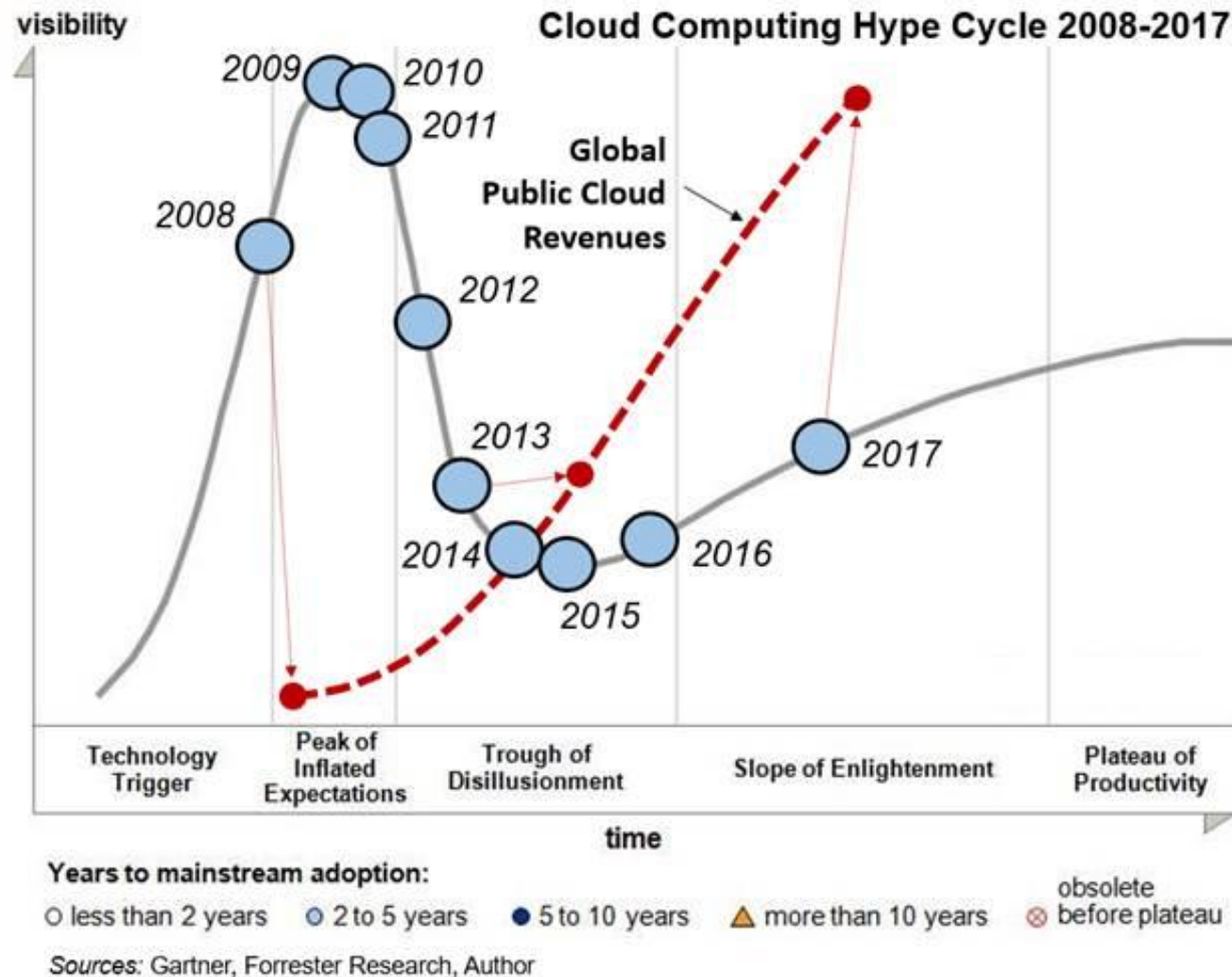
Fundamentals – History



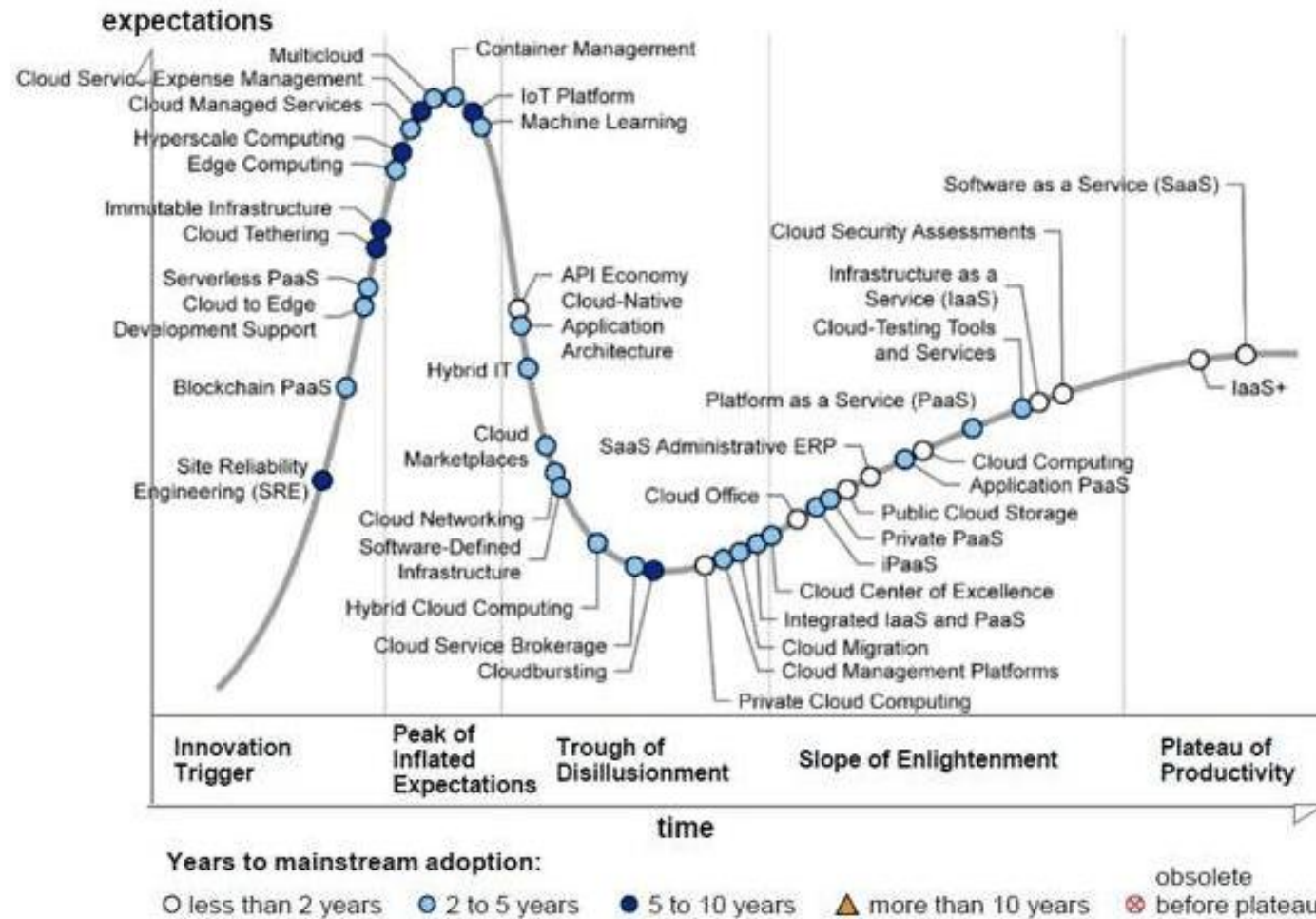
Fundamentals – Only a Buzzword?



Fundamentals – Hype Cycle



Fundamentals – Hype Cycle

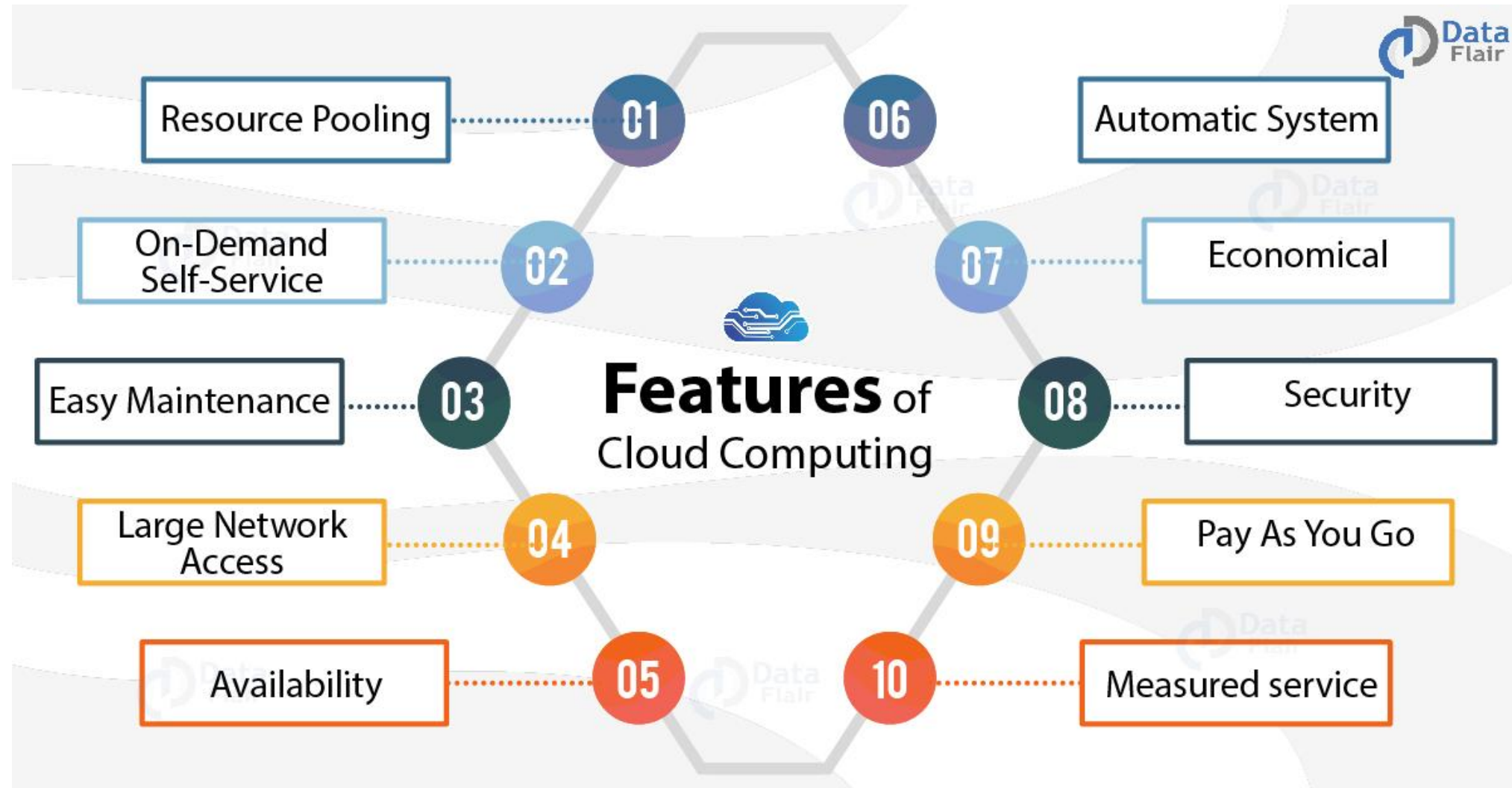


Fundamentals – Essential Characteristics (NIST)

Following National Institute of Standards and Technology (NIST), a cloud system has the following five characteristics:

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity
5. Measured service

Fundamentals – Essential Characteristics (someone else)



Fundamentals – Advantages of Cloud Computing



In addition:

Fewer maintenance issues, instant software updates, backup and recovery, performance and scaling, massive amount of storage, ... and many more

Fundamentals – Disadvantages of Cloud Computing



In addition:

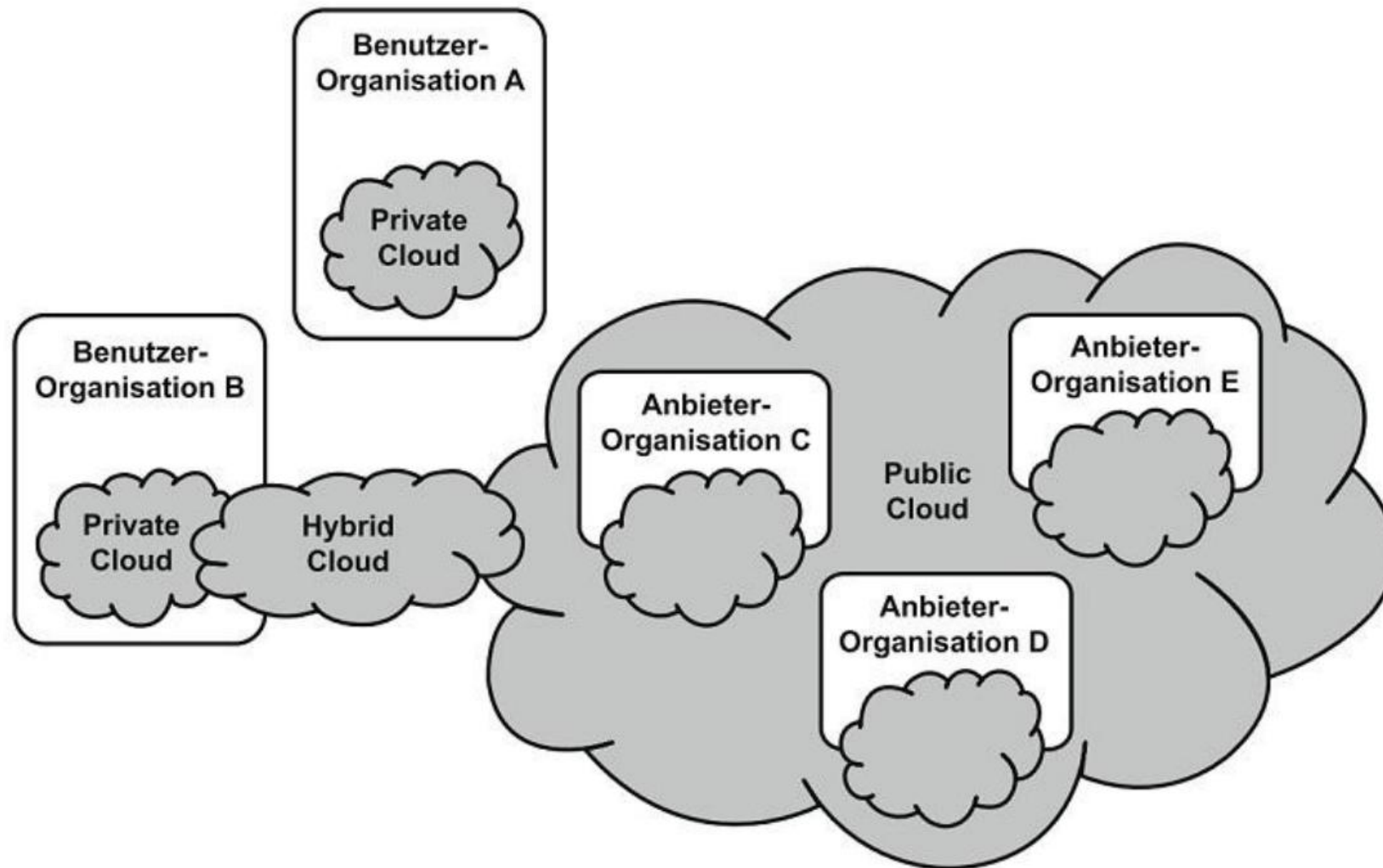
Vendor lock, special requirements, regulatory issues, skill of people (Full Stack), change in the organization, poor quality of the service (sometimes the consumer is the beta tester), ... and many more

Fundamentals – Deployment Models



- **Private** Cloud: Resources of the Cloud are deployed for one particular organization.
- **Community** Cloud: Resources are provided for a community or a group of organizations.
- **Public** Cloud: Resources are provided for everyone in the public.
- **Hybrid** Cloud: Resources are bound together by using different of the other cloud types.
- Enterprise-, IaaS-, IoT-, Distributed-, X-Cloud, ... and many more

Fundamentals – Deployment Models Relationship



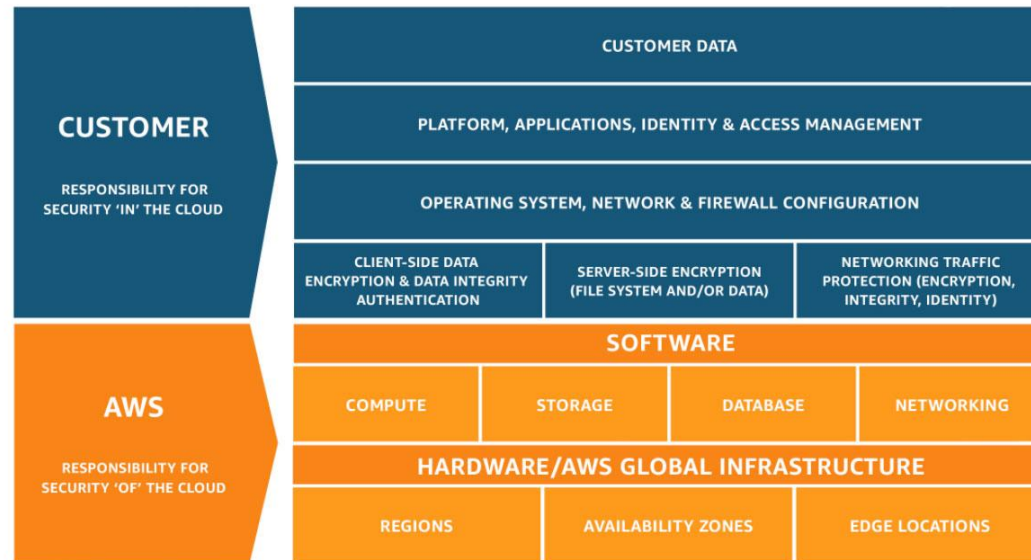
Fundamentals – Deployment Models Relationship

What do you think how a community cloud looks like?

Fundamentals – Shared Responsibility Model for Cloud Security

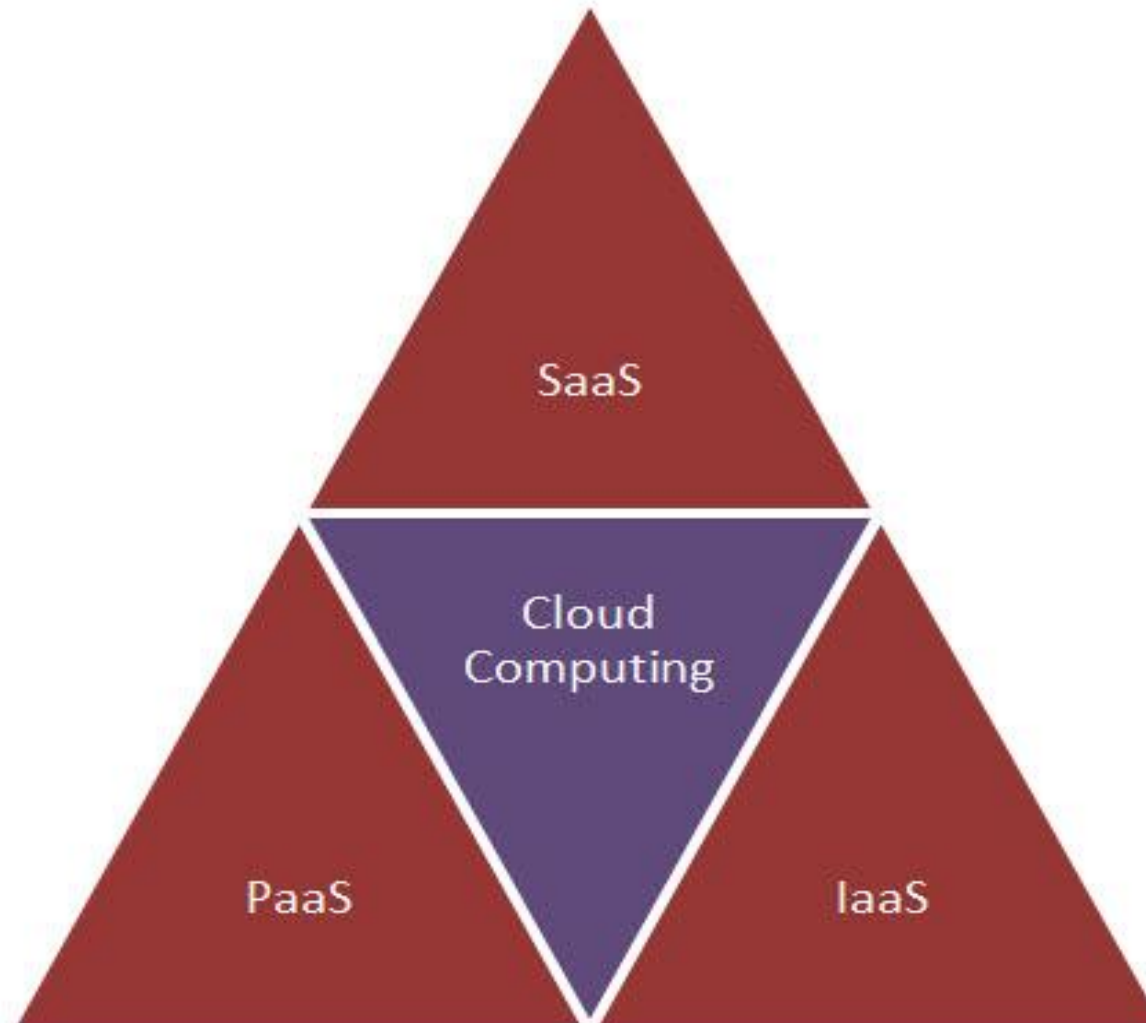
- Security and Compliance is a shared responsibility between the provider and the consumer.
- Depending on the model (IaaS, PaaS, SaaS) the provider and the consumer have different responsibilities.
- The provider is responsible for the security **of** the cloud, while the consumer is responsible for security **in** the cloud.

Fundamentals – Shared Responsibility Model for Cloud Security

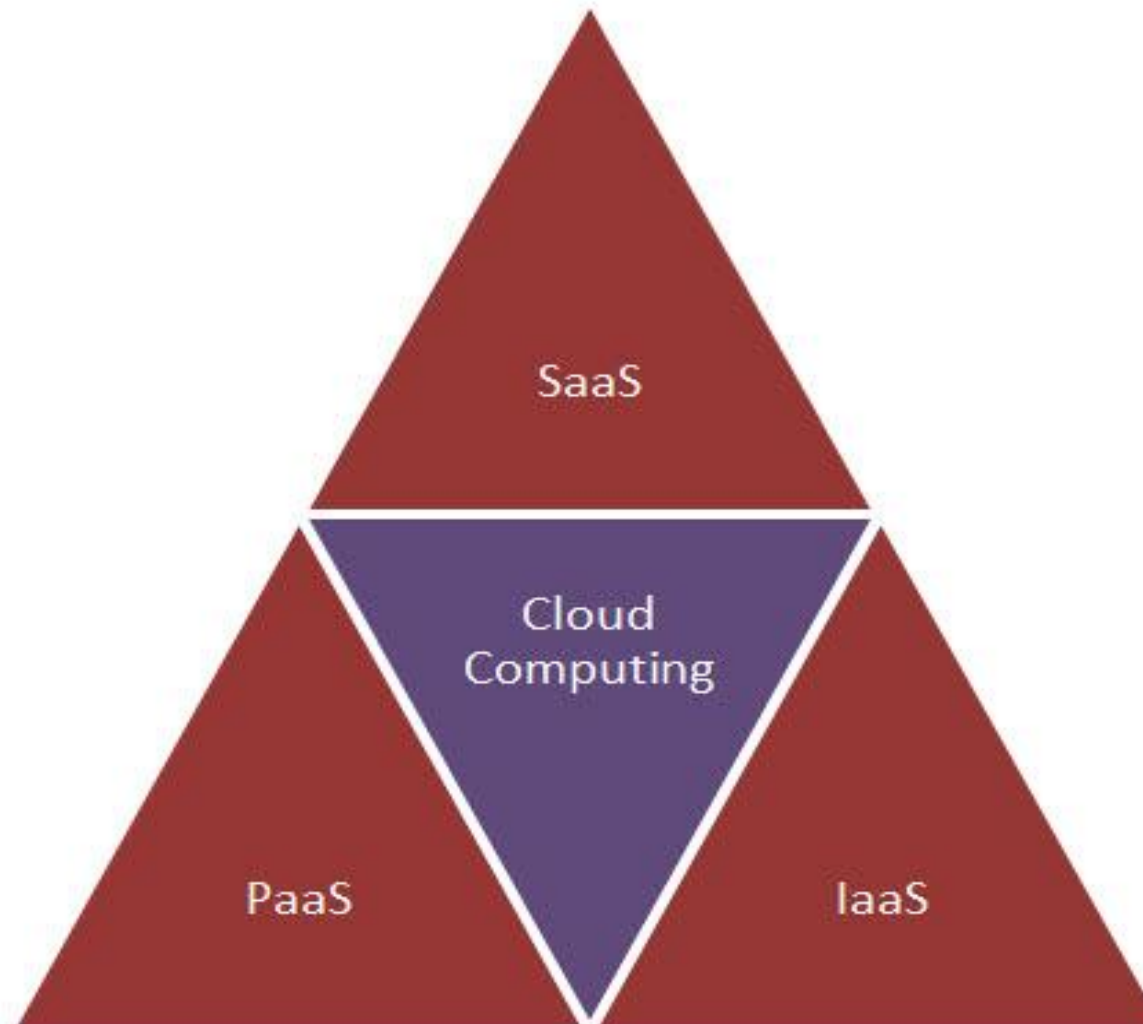


- AWS is responsible for protecting the infrastructure that runs all **of** the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities.
- To ensure security **within** a given cloud environment, the consumer configures and manages the security controls of the selected cloud services (e.g. EC2, S3, RDS, ...).

Fundamentals – Service Models Overview

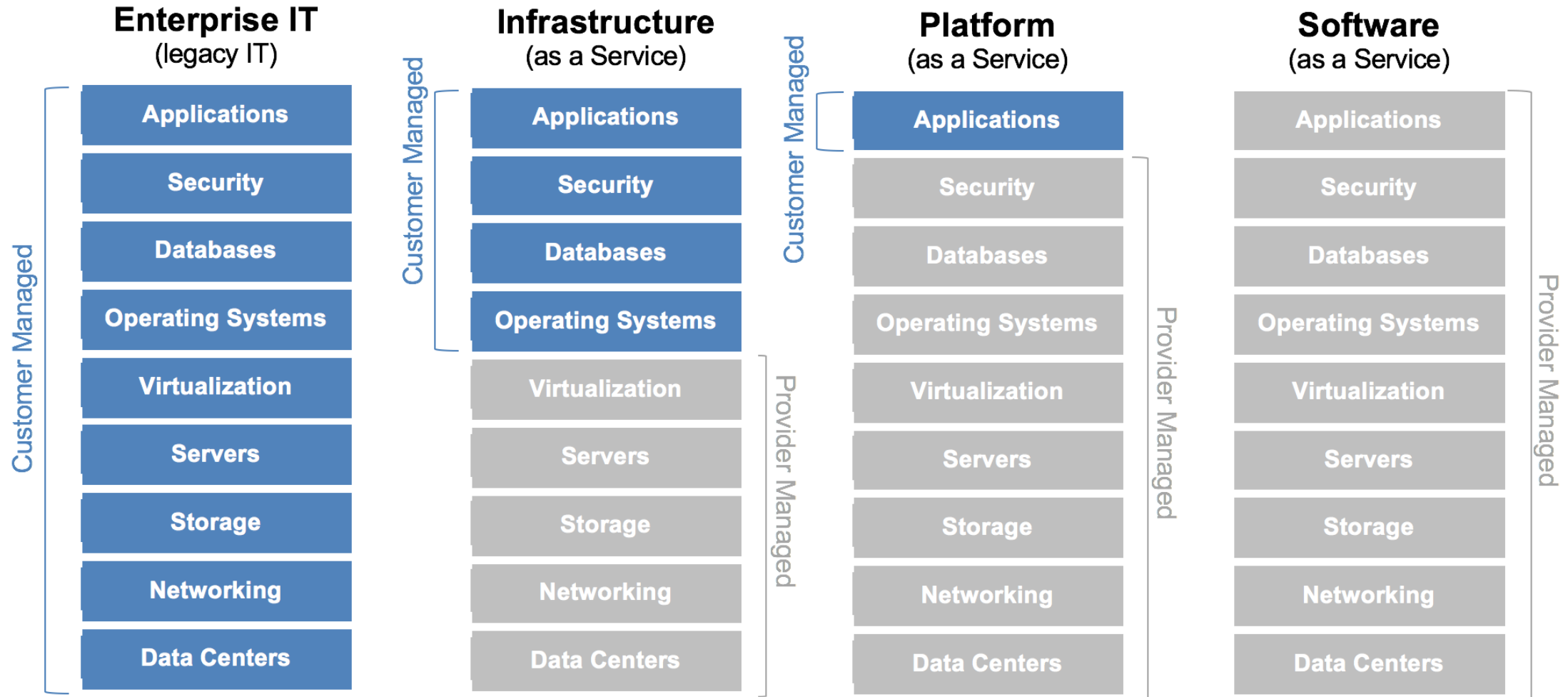


Fundamentals – Service Models Overview

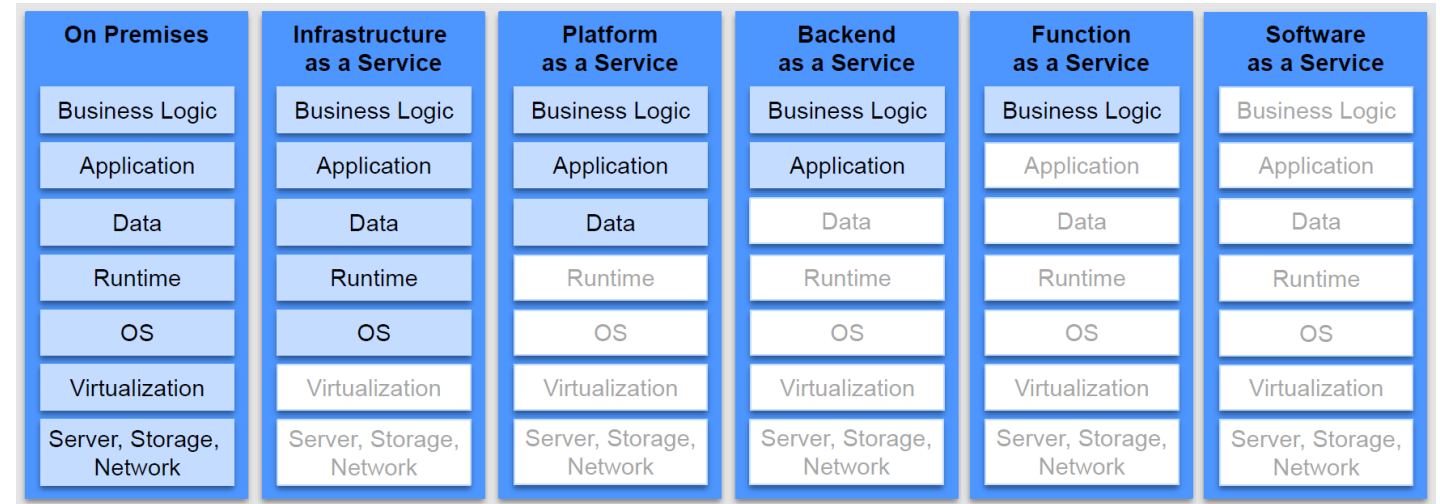
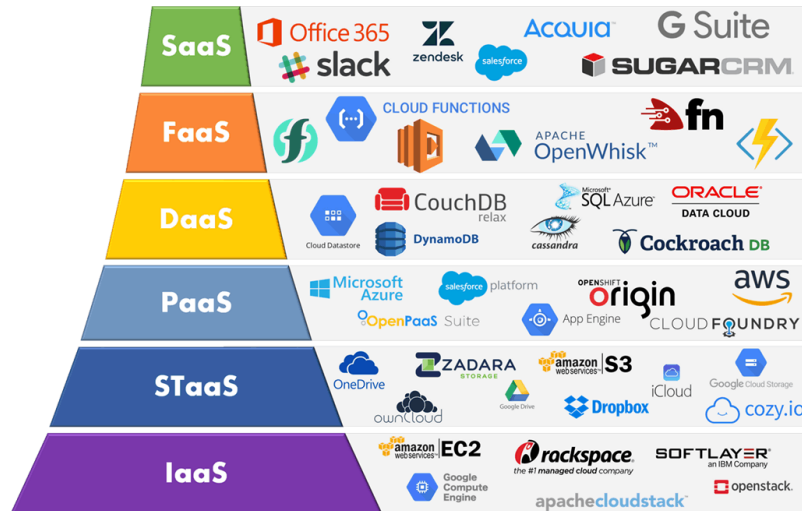


Name a few SaaS, PaaS, IaaS services you know from your daily usage.

Fundamentals – Service Models and Responsibility

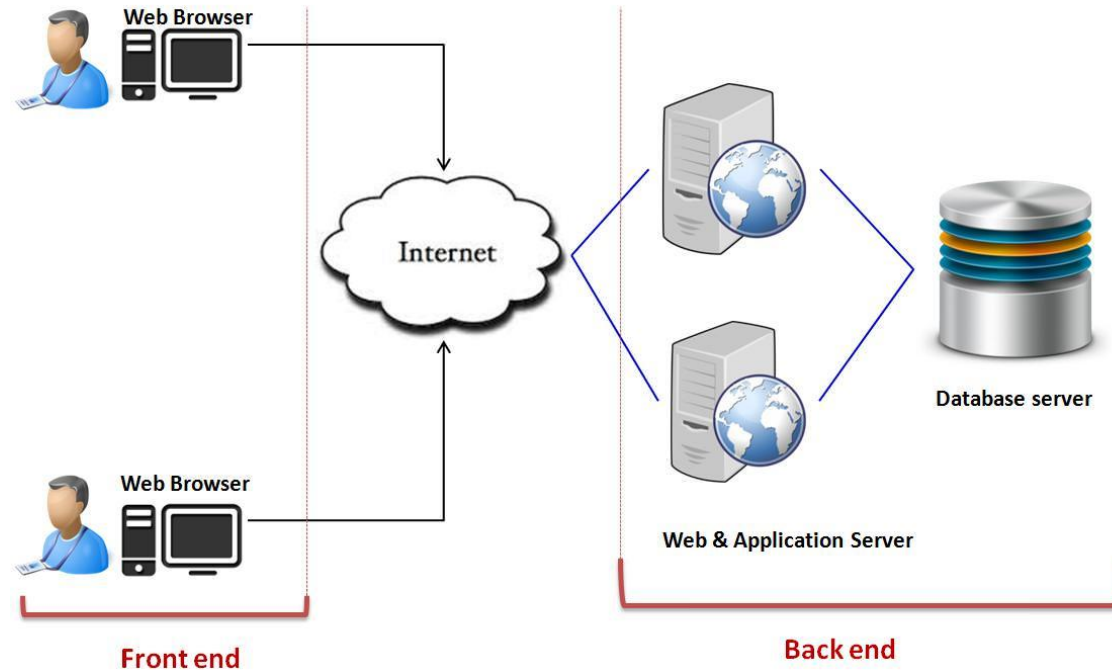


Fundamentals – Service Models and Responsibility



- In practice every vendor has unique offerings.
- Integration capabilities is key for next generation application development.
- Trend is shifting to **XaaS** where X denotes everything.

Fundamentals – Basic Architecture of a Cloud Platform



- **Front end** consist client part of cloud computing system. It comprise of interfaces and applications that are required to access the cloud computing platform (CLI, UI, SDK, ...).
- **Back end** refers to the cloud itself, it comprises of the resources that are required for cloud computing services (e.g. VMs, server, storage, ...).

Fundamentals – Cloud Platform Administration Architecture



- Communication between client and cloud platform management over **public** Internet.
 - Client **authenticates** against access management of the cloud provider.
 - If access is provided, resources and services can be **administered**.
- A cloud platform is just another software system which can be used to fulfill requirements

Fundamentals – Access Security in a Nutshell (1/2)

- Traditional datacenters have been secured by perimeter security
 - Perimeter-based security intrinsically trusts everyone inside the perimeter.
 - Assuming that if an address originates from a “trusted” network segment, the communication and actions were free of malicious intent.
 - Works for enterprise datacenter where you know and control the entire network in detail.
- How perimeter security fails today
 - Heterogeneous environments have more work and less security
 - Virtualization concentrates risk
 - It's impossible to protect what you don't know
 - ... and many of such new threads will rise up soon

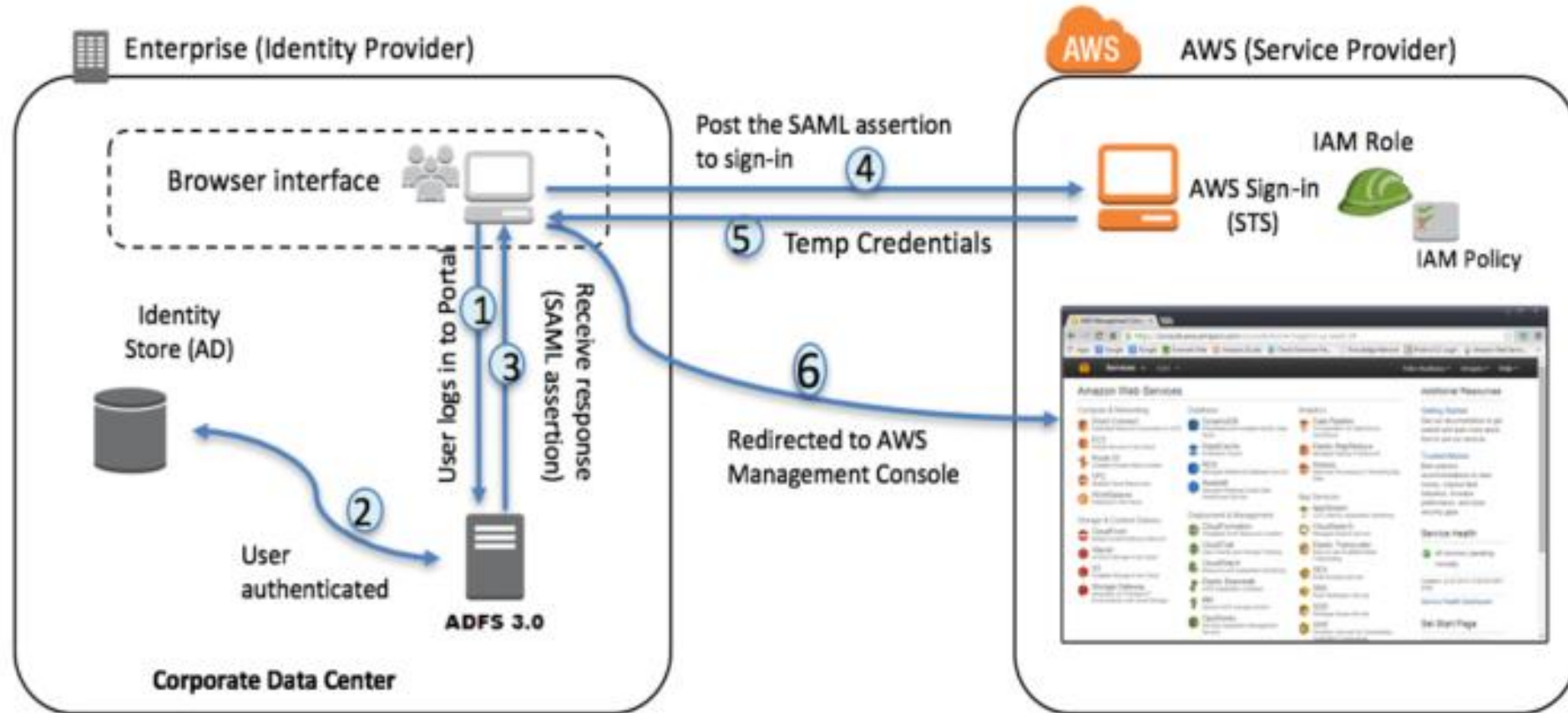
→ Since in today's world it's all about integration, perimeter security is no longer efficient.

→ Zero Trust Model can help to gain a more secure environment.

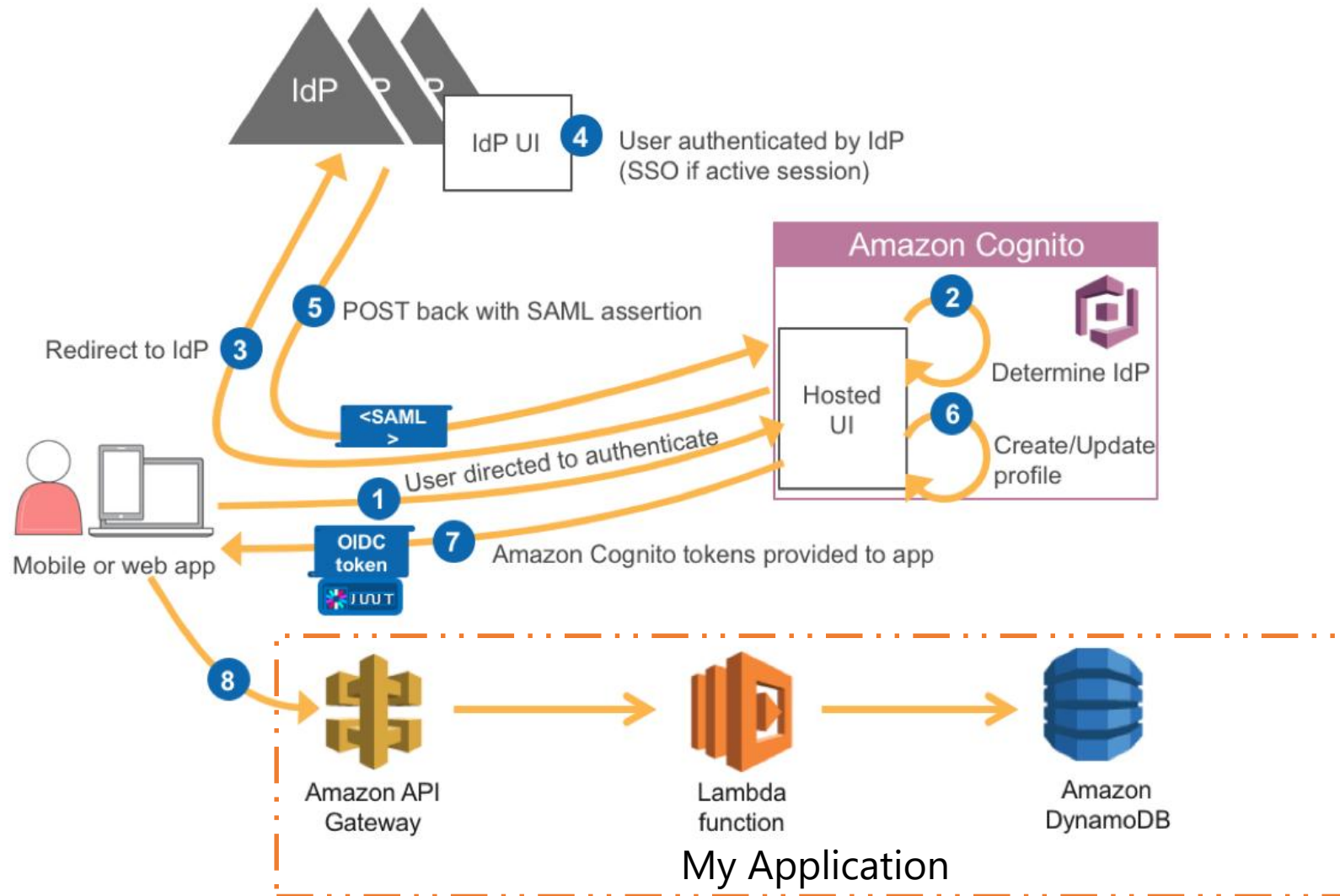
Fundamentals – Access Security in a Nutshell (2/2)

- Since digital transformation is set as key to the future of business, integration and heterogeneous systems become more important.
- Zero Trust Model can help to tackle that challenges
 - Basic principal: „never trust, always verify“.
 - The network is by default a hostile place consisting of internal and external threads.
- This approach shift dictates that trust is only established when the **secure identity** of applications, users and hosts are **authenticated**.
- Controlling access by zero trust allows to manage access independent of the network or the perimeter.

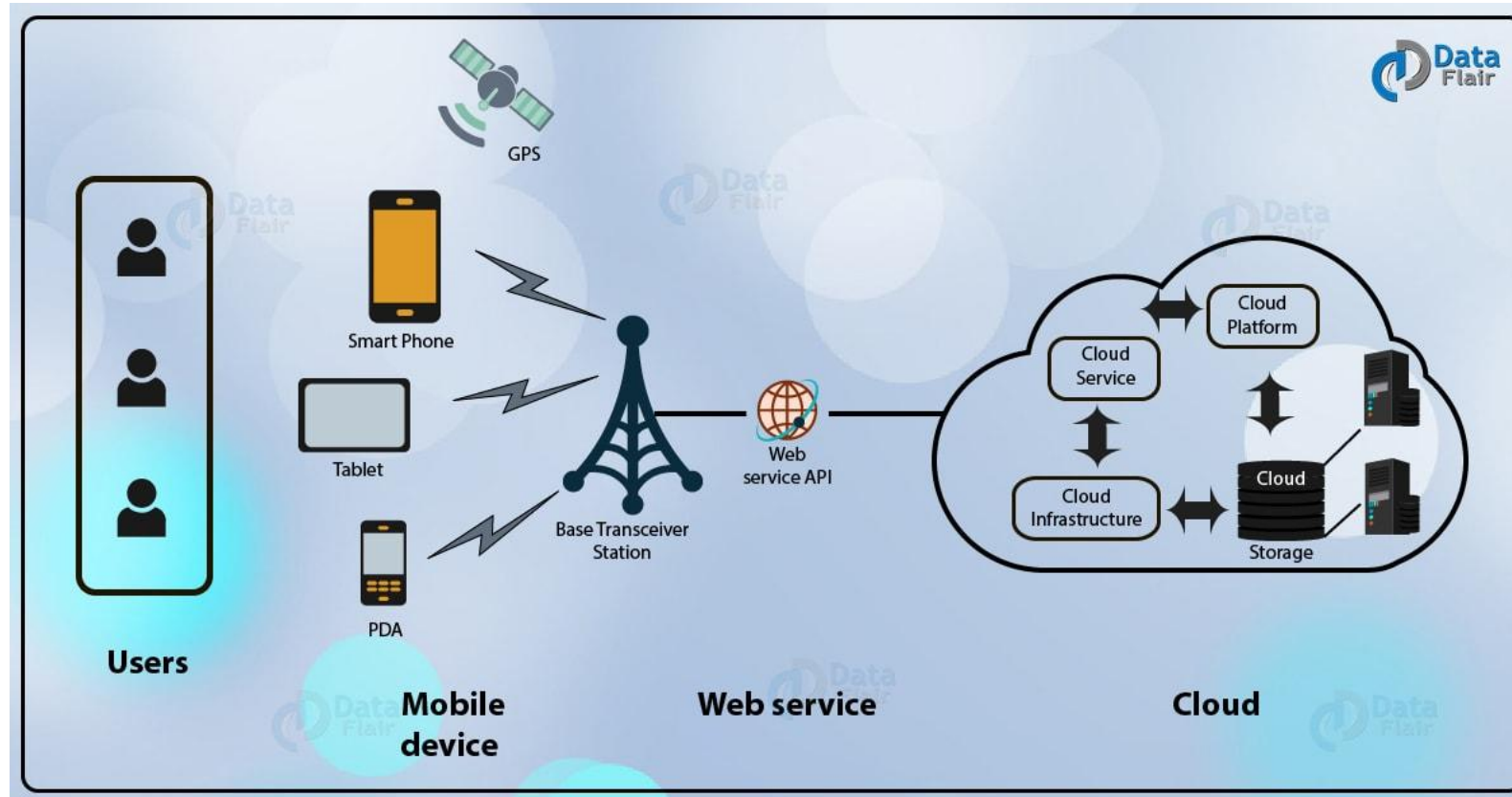
Fundamentals – Access Control Example AWS Web Console



Fundamentals – Access Control Example AWS Web Application



Mobile Cloud Computing in a Nutshell



- In a mobile scenario, only the access path to the Cloud provider is different.
- In best case, all services are device independent.



Cloud Computing

Vorlesung an der Hochschule Karlsruhe -
Technik und Wirtschaft im
Wintersemester 2019/20

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Die Vorlesung im Web:

<http://www.home.hs-karlsruhe.de/~mage0003/>



(Load) Scaling Einführung

- Case-Study:
„Wachstum eines Web-Services“
- Was machen eigentlich...
 - Facebook
 - Netflix
 - ...





Motivation

- Erfolg:
 - Slashdot Effekt → führ zu unvorhersehbarem Traffic auf Ihrer Webseite
- Problem Komplexität
 - Wissenschaftsbereich (CERN, ...)



Tag 1: im Leben eines Web Services

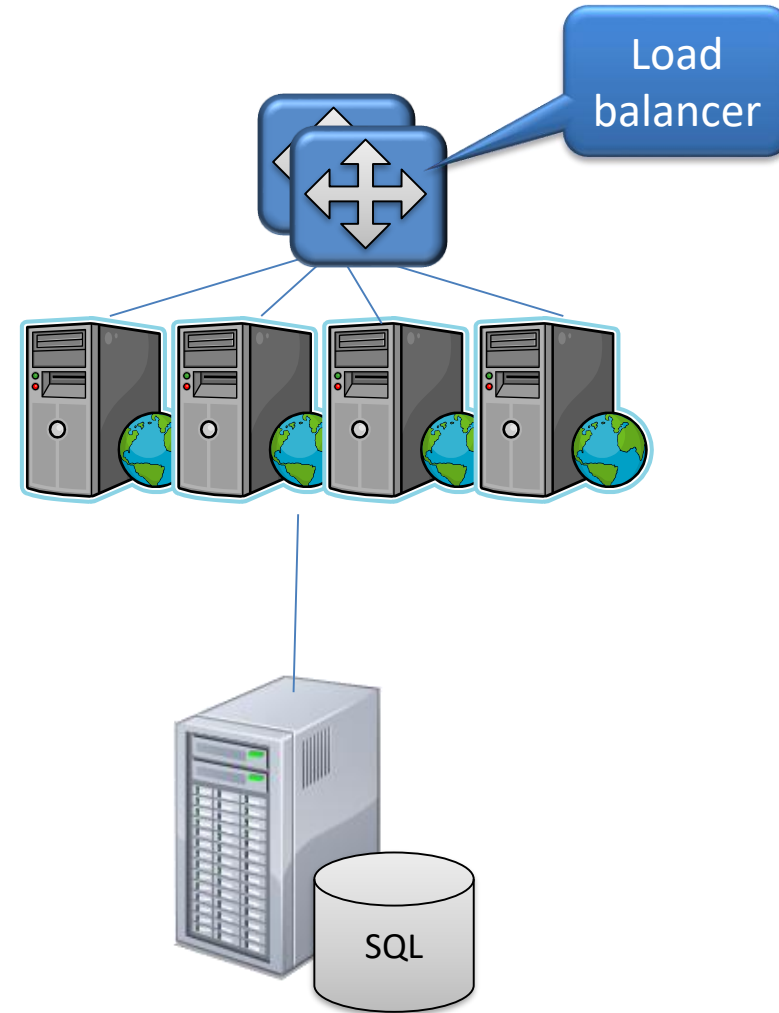
- Stats:
 - 40.000 PI/Tag
 - 0,5 Requests / s
 - ~2-3 Requests / s Peak
- Architektur:
 - Single Server
 - LAMP
 - Linux
 - Apache
 - MySQL
 - PHP





Tag 2: im Leben eines Web Services

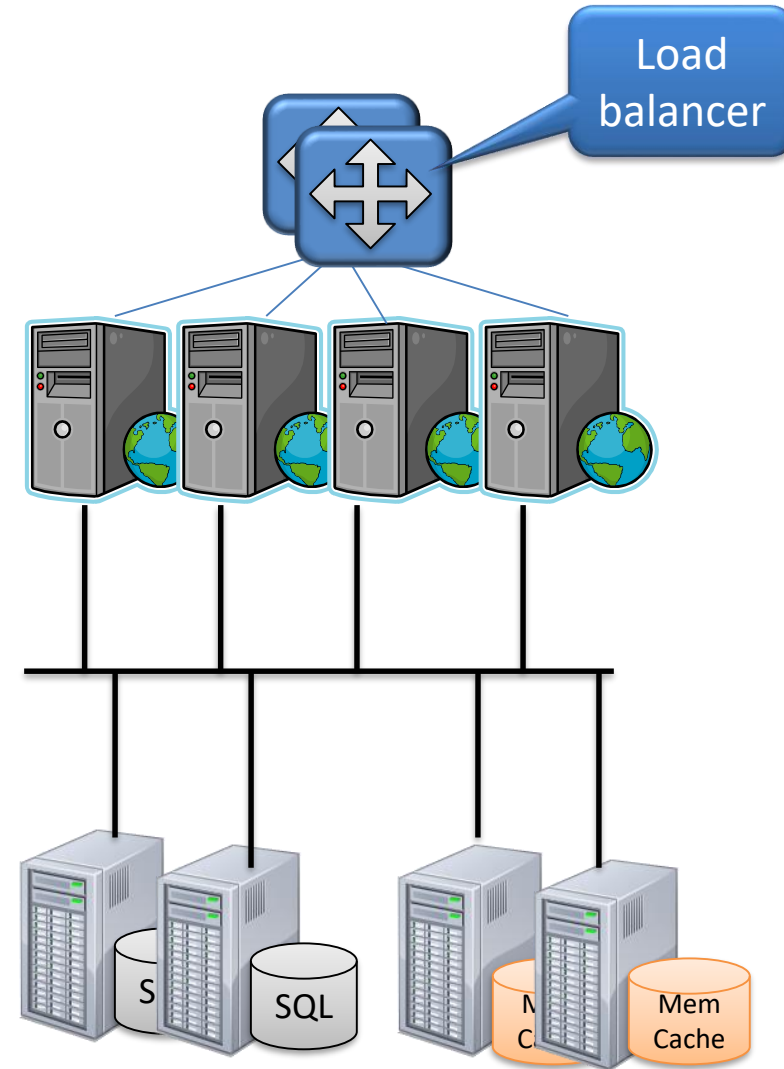
- Stats:
 - 2.000.000 PI/Tag
 - ~20 Requests / s
 - ~100 Requests / s Peak
- Architektur:
 - ~1-10 Server
 - Load Balancer
 - Webserver Farm
 - Apache
 - PHP
 - MySQL





Tag 3: im Leben eines Web Services

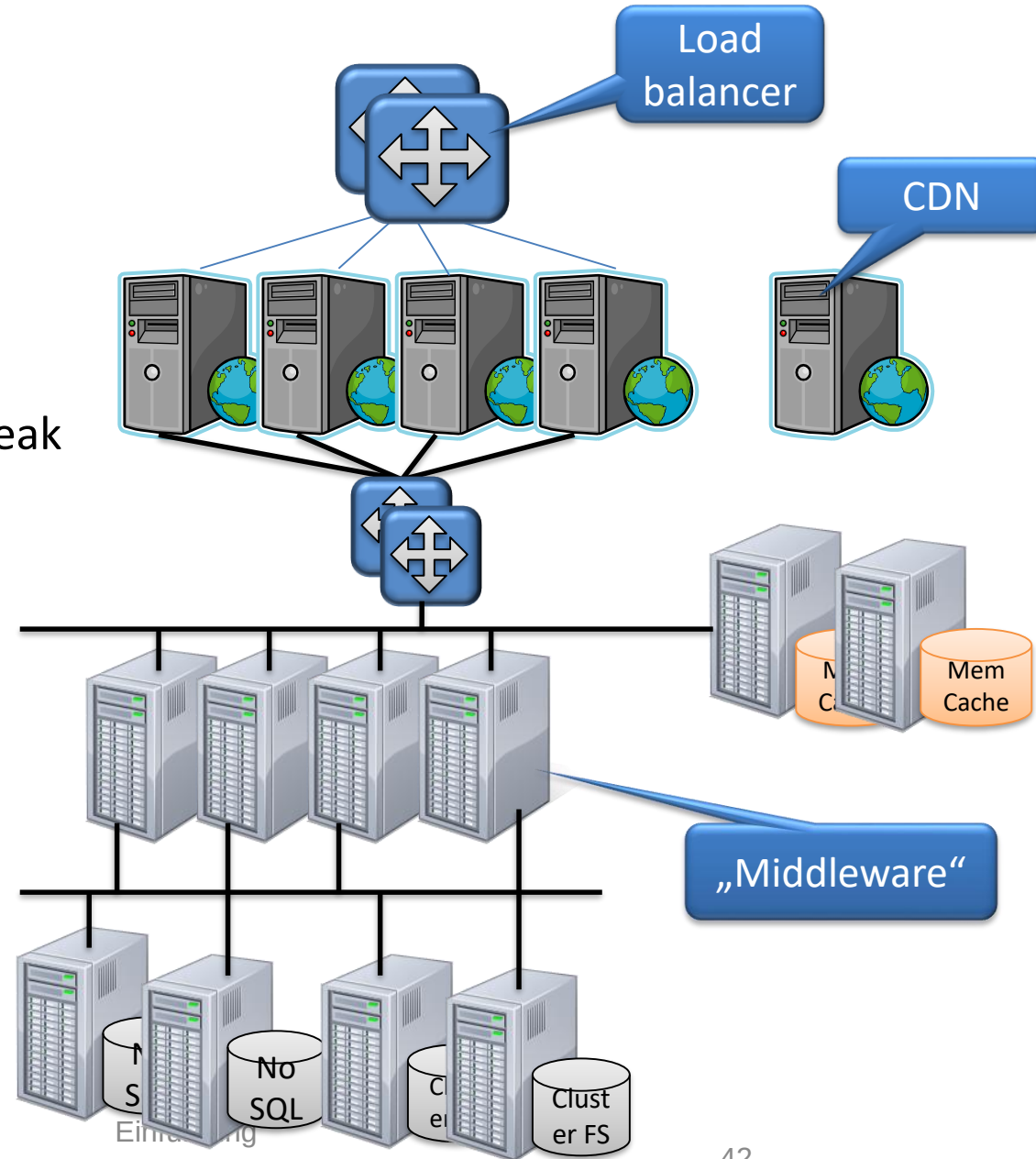
- Stats:
 - 10.000.000 PI/Tag
 - ~100 Requests / s
 - ~500 Requests / s Peak
- Architektur:
 - ~10-100 Server
 - MySQL Master Master Replikation, Read Slaves
 - Memcached
 - Änderung der Applikation!
 - Spezialisierung der Use-Cases



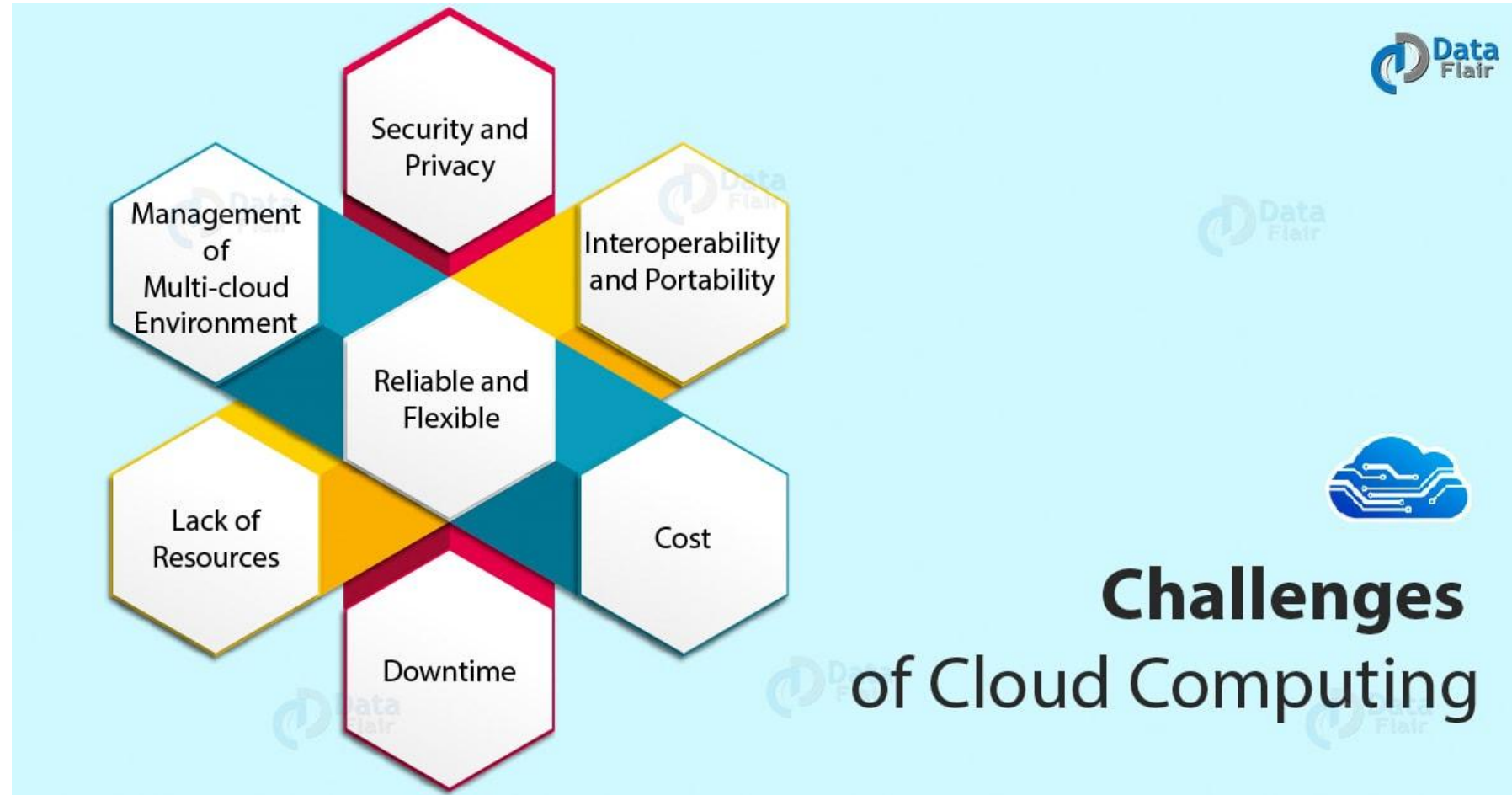


Tag 4: im Leben eines Web Services

- Stats:
 - 1.000.000.000 PI/Tag
 - ~10000 Requests / s
 - ~50000 Requests / s Peak
- Architektur:
 - ~100-500 Server
 - „NoSql“
 - N-Tier Architektur
 - Optimiertes Caching
 - Content Delivery Networks



Challenges of Cloud Computing



- Besides technology, organizational challenges, methodology and skills are also hard to solve ...

Cloud Service Providers

Figure 1. Magic Quadrant for Cloud Infrastructure as a Service, Worldwide



Source: Gartner (July 2019)

Known other providers

- Amazon Web Service (AWS)
- Microsoft Azure
- Google Cloud Platform
- IBM Cloud Services
- Adobe Creative Cloud
- Kamatera
- VMware
- Rackspace
- Red Hat
- Salesforce
- Oracle Cloud
- SAP
- Verizon Cloud
- Navisite
- Dropbox